Shanmugam Subramanian

Mechanical Engineer II | CAE & Structural Simulation | Hyderabad, India

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SUMMARY

Mechanical engineer with 4 years of NPD and NPI experience in antenna and radar hardware, specializing in FEA and CFD. Ran harmonic, random vibration, static and non-linear structural analyses (80% first-pass test correlation) and Ansys Fluent design-of-experiments on 135+ radome profiles (12–18% wind load reduction). Experienced in design optimization, hand-calculation validation and documenting simulation workflows for team adoption.

EXPERIENCE

JMA Wireless — Hyderabad, India
Mechanical Engineer II May 2023 – Present

CAE, Simulation & CFD

- Performed **harmonic & random vibration analyses** (80% first-pass test correlation, IEC 60068-2-6/64) & **IEC 60068-2-31 drop tests** for antenna qualification.
- Ran **non-linear FEA** of a production snap-fit: predicted 38 N insertion & 30 N removal force; flagged a fillet redesign.
- **Ansys Fluent CFD** radome study (2D/3D): DOE on 135+ profiles gave **12–18% wind load reduction**; **patent** in progress.
- Built Excel-based **hand-calculation tools** for drag force, pneumatic crimping force, & plastic snap-joint behavior.
- Automated **Ansys Mechanical pre-processing** with Python/ACT scripts (meshing at coarse/medium/fine profiles, material assignment, midsurfacing of thin-walled bodies, boundary condition & solver setup), **cutting setup time by $\sim$2 hrs per simulation job** & standardizing models across the team.

Mechanical Design

- Applied **topology optimization** to FDM 3D-printed prototypes, **cutting 60 g per unit** at required stiffness.
- Drove **simulation-led design** (snap-fit, topology, FEA) for 3 molded parts & 1 FRP radome: **first-pass success**, **$\sim$$15K under budget**.

Product Development & NPI

- Owned end-to-end **NPD & NPI** for eight 5G Macro & Distributed Antenna Systems products for the US & Canada markets.
- Released RF PCB, sheet metal, die-cast, CNC, & EDM parts with a **$>$90% FAI pass rate**.
- Built RF PCB fixtures, press-pull tools, & alignment jigs: **40% less rework, 75% faster riveting**, 20–50 min saved per unit.
- Managed **ECRs, FAIs, & SVPRs** in Salesforce Propel **PLM** & SolidWorks **PDM** with the global team.
- Supported **contract manufacturing** setup in Chennai through BOM costing, tooling transfer, & operator training.

MulticoreWare India Pvt. Ltd. (Uhnder Team) — Chennai, India
Mechanical Design Engineer (R&D) Jul 2022 – Apr 2023

- Enhanced chassis thermal properties by refining fin geometry, reducing **internal temperature by 6°C**.
- Developed two automotive radar antenna chassis and radomes with **IP68 (IEC 60529)** ratings for ADAS applications.
- Lowered radome **cost by 8%** through material optimization without degrading dielectric performance.

MulticoreWare India Pvt. Ltd. (Uhnder Team) — Chennai, India
Mechanical Project Intern Mar 2022 – Jun 2022

- Updated automotive radar housing designs based on lead engineer feedback & prepared GD&T-compliant drawings.
- Implemented **snap-fit** radome design, reducing the radar BOM by two components and saving $0.3 per unit.

EDUCATION

Bachelor of Engineering — Mechanical Engineering GPA: 8.21 / 10
Alagappa Chettiar College of Engineering and Technology (Anna University) 2018 – 2022

PROJECTS AND COMPETITIONS

Venturi Fan Blade Mechanism Patent in progress
CFD-validated ducted fan blade concept for secondary jet flow augmentation; patent filing in progress.

FDM Subsonic Wind Tunnel Personal project
Designed and 3D-printed a functional subsonic wind tunnel for aerodynamic testing of 1:16 scale die-cast cars.

SAE Bicycle Design Competition — 2020 - 2021 SAE INDIA
Developed a planetary gear-based bicycle drivetrain achieving 15% higher torque than a typical bicycle.

SKILLS & SOFTWARE

Design: SolidWorks, AutoCAD, Fusion 360, Inventor, Onshape, GD&T, Tolerance stackup, 3D Printing
Simulation: **Ansys Mechanical** (static, non-linear, harmonic, random vibration), **Ansys Fluent** (2D/3D CFD), FEA, topology optimization, DOE, hand-calculation tools
Programming & Embedded systems: Python, MATLAB Simulink & Simscape, C, C++, Arduino
Other Tools: Excel, PowerPoint, PLM (Salesforce Propel), SolidWorks PDM
Standards & Testing: IEC 60068-2-6/-64/-31, FOS-driven design, CAE-to-test correlation
Management & Quality: RCA, Risk Mitigation, Feasibility Analysis, Cross-functional Team Coordination